

7.2 Teichmüller's Theorem

Thm 7.2.1 If X is a finite type Riemann surface then $F: B(X) \rightarrow \mathcal{T}_X$ is a homeomorphism.

Open unit ball in $Q'(X)$

Pf: $\forall q \in Q'(X)$ nonzero $\frac{q}{\|q\|}$ is an infinitesimal Beltrami form with $L^\infty = 1$

For $q \in B(X)$ ($\|q\| < 1$) consider $\mu = \|q\| \frac{\bar{q}}{\|q\|}$

Solve corresponding Beltrami eq. get a new complex structure $\rightarrow$ Riemann surface X_q

In X_q , analytic fun ζ satisfies $\bar{\partial}\zeta = \mu \partial\zeta = \|q\| \frac{\bar{q}}{\|q\|} \partial\zeta$

Let $F(q) = [X_q, \text{id}]$

If $M(X)$ is given L^1 norm $q \mapsto \|q\|_{L^1}$ continuous.

$f^\mu: X \rightarrow X_\mu$ is solution $\therefore \mu \mapsto f^\mu$ is continuous. (Ahlfors-Bers)

$\Rightarrow F$ is continuous.

Then if $F(q_1) = F(q_2)$, then, $\exists \alpha: X_{q_1} \rightarrow X_{q_2}$ s.t. α is analytic homeomorphic

homotopic to id.

Consider $\text{id}: X \rightarrow X_{q_1}$, $\text{id}: X \rightarrow X_{q_2}$, they are Teichmüller maps.

7.2 Teichmüller's Theorem

Thm 7.2.1 If X is a finite type Riemann surface
then $F: B(X) \rightarrow T_X$ is a homeomorphism.

Cor 7.2.3
 $f: X \rightarrow X$

$\uparrow$
Open unit ball in $Q^1(X)$

$$\text{If } d(F(q_n), F(q_0)) \leq M \quad \ln\left(\frac{1 + \|q_n\|}{1 - \|q_n\|}\right) \leq M$$
$$\|q_n\| \leq \frac{e^{M-1}}{e^M + 1} \Rightarrow \text{cpt} \Rightarrow \text{proper}$$

$\therefore F$ is a homeomorphism between two spaces.



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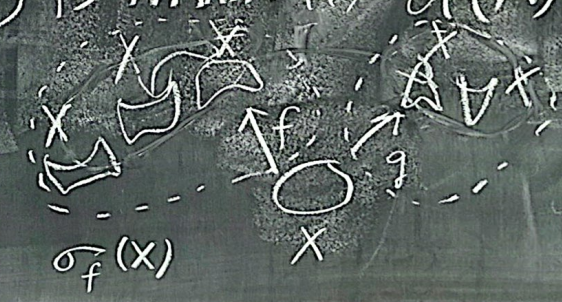
Cor 7.23. X : Riemann surface of finite type.
 $f: X \rightarrow X$ is a homeomorphism. Define $G_f(x) \in \overline{T}_x$ is
 $(X, f: x \rightarrow x)$. Then $\exists!$ Teichmüller mapping $g: X \rightarrow X$
homotopic to f , and $d(X, G_f(x)) = \ln K(g)$

Pf: $\exists q \in B(X)$, s.t $F(q) = G_f(x)$, $F(q) = (X_q, id)$ $\therefore \exists$ conformal

$\alpha: X_q \rightarrow X$ s.t $\alpha \circ id \sim f$

def $g = id: X \rightarrow X_q$ $g \sim f$

By 53.8, $K(g)$ is minimal, $d(X, G_f(x)) = \inf_{\phi \sim f} \ln K(\phi) = \ln K(g)$



$$L(X) = \frac{(\xi' - \Lambda_2)}{T} \circ \partial \bar{\partial}^{-1}$$

$$p: D \rightarrow \boxed{p^{-1}(x)}$$